

1. The leading order Feynman graphs contributing to the $\psi\psi \rightarrow \psi\psi$ scattering are shown in Fig. 1. These graphs differ by a relative sign due to the permutation of the external lines:

$$i\mathcal{T}_{\psi\psi \rightarrow \psi\psi} = i\mathcal{T}_t - i\mathcal{T}_u$$



Figure 1: Feynman graphs for $\psi\psi \rightarrow \psi\psi$ scattering

The interaction vertex is equal to

$$g_1 \mathbb{1} + ig_2 \gamma^5 =: \alpha$$

By reading out the Feynman graphs, we obtain the following relations for the amplitudes:

$$i\mathcal{T}_t = \bar{u}(k_1, \sigma_1) \alpha u(p_1, s_1) \frac{1}{t - M^2 + i0} \bar{u}(k_2, \sigma_2) \alpha u(p_2, s_2)$$

$$i\mathcal{T}_u = \bar{u}(k_1, \sigma_1) \alpha u(p_2, s_2) \frac{1}{u - M^2 + i0} \bar{u}(k_2, \sigma_2) \alpha u(p_1, s_1)$$

I assume that the final result for the T-matrix should be obtained for an unpolarized scattering process and therefore I implicitly average over all initial and final spin states.

$$|\mathcal{T}_{\psi\psi \rightarrow \psi\psi}|^2 = \frac{1}{4} \sum_{s_1, s_2, \sigma_1, \sigma_2} \left(\left[\bar{u}(k_1, \sigma_1) \alpha u(p_1, s_1) \frac{1}{t - M^2} \bar{u}(k_2, \sigma_2) \alpha u(p_2, s_2) \right. \right. \\ \left. \left. - \bar{u}(k_1, \sigma_1) \alpha u(p_2, s_2) \frac{1}{u - M^2} \bar{u}(k_2, \sigma_2) \alpha u(p_1, s_1) \right] \right. \\ \left. \left[\bar{u}(p_2, s_2) \alpha u(k_2, \sigma_2) \frac{1}{t - M^2} \bar{u}(p_1, s_1) \alpha u(k_1, \sigma_1) \right. \right. \\ \left. \left. - \bar{u}(p_1, s_1) \alpha u(k_1, \sigma_1) \frac{1}{u - M^2} \bar{u}(p_2, s_2) \alpha u(k_2, \sigma_2) \right] \right)$$

$$-\bar{u}(p_1, s_1)\alpha u(k_2, \sigma_2)\frac{1}{u-M^2}\bar{u}(p_2, s_2)\alpha u(k_1, \sigma_1)\Bigg)\Bigg),$$

where we have used the fact that

$$\alpha^\dagger = g_1 \mathbb{1} - i\gamma_2 \gamma^5 \implies \gamma^0 \alpha^\dagger \gamma^0 = g_1 \mathbb{1} + i\gamma_2 \gamma^5 = \alpha,$$

where the γ^0 matrices stem from the Dirac conjugation of the wavefunctions.

Now we can reorganize the groups $\bar{u}\alpha u$, as they are scalars, and use the trace trick. Furthermore, after rearranging the terms, we use the polarization sum identity

$$\sum_s u(p, s)\bar{u}(p, s) = \gamma \cdot p + m.$$

Overall, we obtain

$$\begin{aligned} |\mathcal{T}_{\psi\psi\rightarrow\psi\psi}|^2 &= \frac{1}{4} \frac{1}{(t-M^2)^2} \text{Tr}[(\not{k}_1 + m)\alpha(\not{p}_1 + m)\alpha] \text{Tr}[(\not{k}_2 + m)\alpha(\not{p}_2 + m)\alpha] \\ &\quad + \frac{1}{4} \frac{1}{(u-M^2)^2} \text{Tr}[(\not{k}_1 + m)\alpha(\not{p}_2 + m)\alpha] \text{Tr}[(\not{k}_2 + m)\alpha(\not{p}_1 + m)\alpha] \\ &\quad - \frac{1}{4} \frac{1}{(t-M^2)(u-M^2)} \text{Tr}[(\not{k}_1 + m)\alpha(\not{p}_1 + m)\alpha(\not{k}_2 + m)\alpha(\not{p}_2 + m)\alpha] \\ &\quad - \frac{1}{4} \frac{1}{(t-M^2)(u-M^2)} \text{Tr}[(\not{k}_1 + m)\alpha(\not{p}_2 + m)\alpha(\not{k}_2 + m)\alpha(\not{p}_1 + m)\alpha]. \end{aligned}$$

Finally, we have to compute the traces of products of gamma matrices. We employ the following rules

- $\text{Tr}[\gamma^{\mu_1} \dots \gamma^{\mu_n}] = 0$ for odd n
- $\text{Tr}[\gamma^{\mu_1} \dots \gamma^{\mu_n} \gamma^5] = 0$ for odd n
- $\text{Tr}[\gamma^5] = \text{Tr}[\gamma^\mu \gamma^\nu \gamma^5] = 0$
- $\text{Tr}[\gamma^\mu \gamma^\nu] = 4\eta^{\mu\nu}$
- $\text{Tr}[\gamma^\mu \gamma^\nu \gamma^\kappa \gamma^\lambda] = 4(\eta^{\mu\nu}\eta^{\kappa\lambda} - \eta^{\mu\kappa}\eta^{\nu\lambda} + \eta^{\mu\lambda}\eta^{\nu\kappa})$

For the first type of trace we quickly obtain

$$\text{Tr}[(\not{k}_1 + m)\alpha(\not{p}_1 + m)\alpha] = 4[(g_1^2 + g_2^2)(k_1 \cdot p_1) + (g_1^2 - g_2^2)m^2].$$

For the second type trace we get

$$\begin{aligned} \frac{1}{4} \text{Tr} \left[(\not{k}_1 + m) \alpha (\not{p}_1 + m) \alpha (\not{k}_2 + m) \alpha (\not{p}_2 + m) \alpha \right] = \\ = (g_1^2 + g_2^2) [(k_1 \cdot k_2)(p_1 \cdot p_2) - (k_1 \cdot p_2)(k_2 \cdot p_1) - (k_1 \cdot p_1)(k_2 \cdot p_2)] \\ - m^2 [(g_1^2 + g_2^2)^2 (k_1 \cdot k_2 + p_1 \cdot p_2) + (g_1^4 - g_2^4)(k_1 \cdot p_1 + k_1 \cdot p_2 + k_2 \cdot p_1 + k_2 \cdot p_2)] \\ - m^4 (g_1^4 - 6g_2^2 g_1^2 + g_2^4), \end{aligned}$$

where we have used the fact that the traces $\text{Tr}[\gamma^\mu \gamma^\nu \gamma^\kappa \gamma^\lambda \gamma^5]$ have canceled out. Using the symmetry of this result we see that the last trace with $\not{p}_2 \leftrightarrow \not{p}_1$ gives the same contribution to the total squared amplitude.

Overall, we obtain

$$\begin{aligned} |\mathcal{T}|^2 = & \frac{4}{(t - M^2)^2} [(g_1^2 + g_2^2)(k_1 \cdot p_1) + (g_1^2 - g_2^2)m^2] [(g_1^2 + g_2^2)(k_2 \cdot p_2) + (g_1^2 - g_2^2)m^2] \\ & + \frac{4}{(u - M^2)^2} [(g_1^2 + g_2^2)(k_1 \cdot p_2) + (g_1^2 - g_2^2)m^2] [(g_1^2 + g_2^2)(k_2 \cdot p_1) + (g_1^2 - g_2^2)m^2] \\ & - \frac{2}{(t - M^2)(u - M^2)} \left[m^4 (g_1^4 - 6g_2^2 g_1^2 + g_2^4) \right. \\ & \quad (g_1^2 + g_2^2) [(k_1 \cdot p_2)(k_2 \cdot p_1) + (k_1 \cdot p_1)(k_2 \cdot p_2) - (k_1 \cdot k_2)(p_1 \cdot p_2)] \\ & \quad \left. + m^2 [(g_1^2 + g_2^2)^2 (k_1 \cdot k_2 + p_1 \cdot p_2) + (g_1^4 - g_2^4)(k_1 \cdot p_1 + k_1 \cdot p_2 + k_2 \cdot p_1 + k_2 \cdot p_2)] \right] \end{aligned}$$

[2.] The leading order Feynman graphs contributing to the $\bar{\psi}\psi \rightarrow \bar{\psi}\psi$ scattering are shown in Fig. 2. These graphs also differ by a relative sign due to the permutation of the external lines:

$$i\mathcal{T}_{\bar{\psi}\psi \rightarrow \bar{\psi}\psi} = i\mathcal{T}_t - i\mathcal{T}_s$$

The interaction vertex in this case is the same, namely α . By reading out the Feynman graphs, we obtain the amplitudes for the two channels:

$$\begin{aligned} i\mathcal{T}_t = & \bar{u}(k_1, \sigma_1) \alpha u(p_1, s_1) \frac{1}{t - M^2 + i0} \bar{v}(-k_2, \sigma_2) \alpha v(-p_2, s_2) \\ i\mathcal{T}_s = & \bar{v}(-k_2, \sigma_2) \alpha u(p_1, s_1) \frac{1}{s - M^2 + i0} \bar{u}(k_1, \sigma_1) \alpha v(-p_2, s_2) \end{aligned}$$

We use the same procedure to compute the square of the amplitude by averaging over the initial and final spin states as in the previous case. In addition, we use the following polarization sum identity

$$\sum_s v(-p, s) \bar{v}(-p, s) = -\gamma \cdot p - m.$$

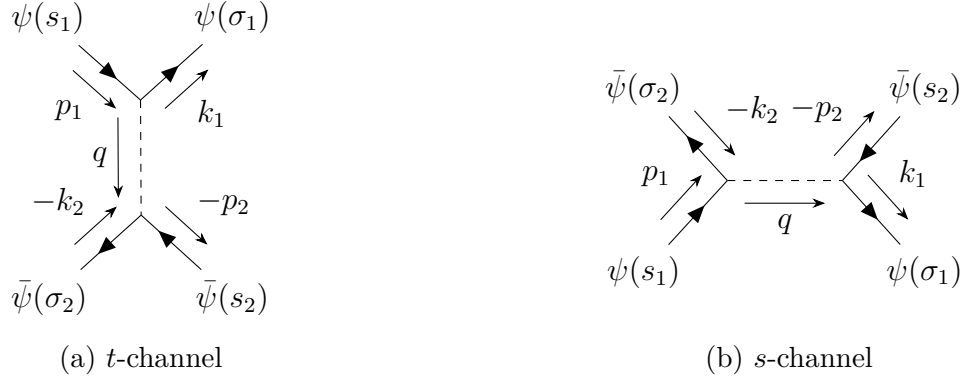


Figure 2: Feynman graphs for $\bar{\psi}\psi \rightarrow \bar{\psi}\psi$ scattering

We can clearly see that in the relation for $|\mathcal{T}_{\psi\psi \rightarrow \psi\psi}|^2$ there are always the polarization sums, which now correspond to antiparticles, in pairs. Therefore, the changes in signs cancel each other and we obtain exactly the same result as for $|\mathcal{T}_{\psi\psi \rightarrow \psi\psi}|^2$

$$|\mathcal{T}_{\psi\psi \rightarrow \psi\psi}|^2(p_1, p_2 | k_1, k_2) = |\mathcal{T}_{\bar{\psi}\bar{\psi} \rightarrow \bar{\psi}\bar{\psi}}|^2(p_1, -k_2 | k_1, -p_2)$$

with the exception that now $u_{\psi\psi \rightarrow \psi\psi} = p_1 - k_2 \equiv p_1 - k_2 = s_{\bar{\psi}\bar{\psi} \rightarrow \bar{\psi}\bar{\psi}}$.

3. We can see that the squared amplitude of the two distinct processes is the same after we correlate

$$\begin{aligned} \text{in: } \psi(p_1, s_1) &\leftrightarrow \text{in: } \psi(k_1, s_1) \\ \text{in: } \psi(p_2, s_2) &\leftrightarrow \text{in: } \bar{\psi}(-k_2, \sigma_2) \\ \text{out: } \psi(k_1, \sigma_1) &\leftrightarrow \text{out: } \psi(k_1, \sigma_1) \\ \text{out: } \psi(k_2, \sigma_2) &\leftrightarrow \text{out: } \bar{\psi}(-p_2, s_1) \end{aligned}$$

We can also rewrite the crossing rules in terms of Mandelstam invariants:

$\psi(p_1)\psi(p_2) \rightarrow \psi(k_1)\psi(k_2)$ $s = (p_1 + p_2)^2$ $t = (p_1 - k_1)^2$ $u = (p_1 - k_2)^2$	$\bar{\psi}(-k_2)\psi(p_1) \rightarrow \bar{\psi}(-p_2)\psi(k_1)$ $u = (p_1 + p_2)^2$ $t = (p_1 - k_1)^2$ $s = (p_1 - k_2)^2$
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Overall, we see that the crossing move replaces $s \leftrightarrow u$.